

Week 3 – Risk & Asset Analysis

Building a Resilient Security & Fencing System for a Sustainable Mango Supply Chain

Project Scope: Sustainable Supply Chain for Dried Mango (Burkina Faso → Germany)

Module Responsibility: Security & Fencing System Engineering

Prepared By: Tania | Systems Engineer & Security Consultant

Date & Status: August 2026 | Milestone Deliverable 03 (Approved Preparation)

SYSTEM ARCHITECTURE BOUNDARY & OPERATIONAL PERIMETER

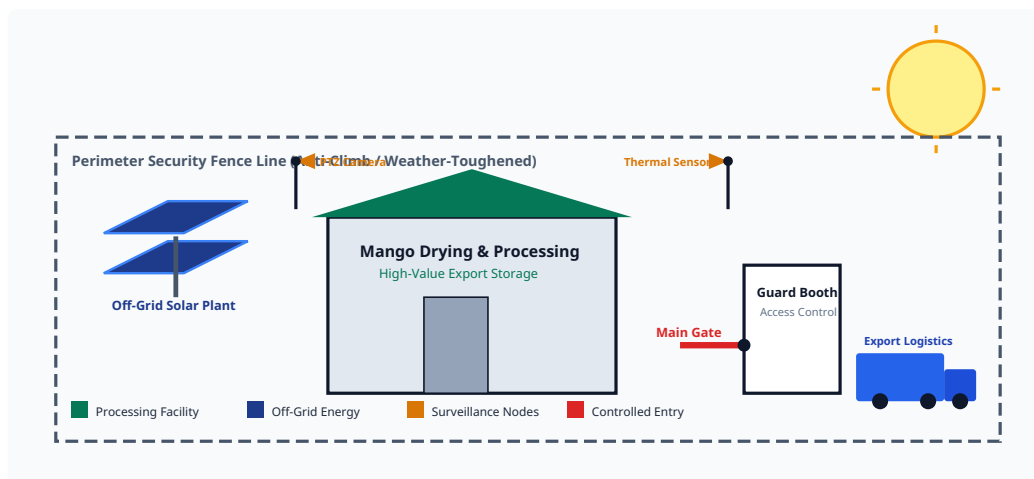


Figure 0.1: Conceptual boundary diagram of the rural organic mango processing facility in Burkina Faso.

1. EXECUTIVE INTRODUCTION & APPROACH

A fundamental vulnerability in physical security engineering is the premature selection of hardware—such as cameras, motion sensors, or barbed wire—prior to conducting a rigorous analysis of the operating domain. Security design must never begin by choosing technologies. It must start by understanding the target environment, the specific operational context, and the real-world threat vectors acting upon the system.

"Good engineering begins with understanding the problem before proposing the solution."
— Core Systems Engineering Principle

PURPOSE OF WEEK 3 ANALYSIS

This report documents the systems engineering process carried out during Week 3 for the **Security & Fencing System** module of the Sustainable Mango Supply Chain project (Burkina Faso to Germany). The primary objective is to evaluate the operational landscape to establish a resilient, context-aware foundation before physical deployment begins.

1. Asset Identification

Cataloging all physical, human, operational, and digital assets that require protection across the agricultural and processing site.

2. Operational Context

Evaluating the extreme environmental, structural, and operational conditions unique to rural Burkina Faso.

3. Threat & Risk Classification

Categorizing risks into six distinct domains to prevent single-point security blind spots and misallocations of budget.

4. Engineering Decisions

Mapping identified threats directly into non-negotiable engineering requirements and physical design parameters for Week 4+.

STRATEGIC SUPPLY CHAIN ALIGNMENT

The facility produces high-value dried organic mangoes for export to international markets in Germany. Maintaining an uninterrupted, secure supply chain requires strict compliance with international food safety guidelines and organic traceability standards. A failure in perimeter security does not merely risk physical loss of inventory; it threatens product contamination, loss of organic certification, and supply chain interruption.

Methodological Continuum: Progression Across Weeks

- **Week 1 (Retrospective Analysis):** Analyzed past design flaws, identifying lack of environmental resilience and unmonitored perimeter gaps.
- **Week 2 (Engineering Strategy):** Defined core principles—Off-Grid Autonomy, Environmental Robustness, and Defense-in-Depth.
- **Week 3 (Risk & Asset Analysis):** Rigorous evaluation of environmental constraints, asset criticalities, and priority threats (Current Phase).

2. FARM OPERATING CONTEXT ANALYSIS

Security hardware deployed in high-stress sub-Saharan environments experiences rapid degradation if not specifically selected for rural African operating conditions. The engineering specifications of the Security & Fencing System must explicitly accommodate four interrelated context pillars.

A. Environmental Context (Sub-Saharan Sahelian Climate)

Key Factors: Ambient temperatures exceeding 40°C–45°C, intense UV radiation, fine Harmattan dust clouds (November–March), heavy seasonal monsoon downpours, high winds, and seasonal bushfire hazards.

Engineering Impact: Standard commercial CCTV thermal throttles or fails under extreme heat; solar panels accumulate dust cover reducing charging efficiency by up to 30%; perimeter fencing requires anti-corrosion coating and thermal expansion joints.

B. Infrastructure Context (Rural Standalone Realities)

Key Factors: Complete absence or unreliability of public electrical grid; unstable cellular and internet coverage; remote location far from municipal emergency services.

Engineering Impact: The security architecture must function on 100% off-grid solar power with localized battery storage (minimum 72-hour autonomy). Communication links must utilize localized mesh networks and localized edge storage to prevent data loss during network outages.

C. Operational Context (High-Throughput Harvest Dynamics)

Key Factors: Daily movement of 100+ seasonal agricultural workers; frequent heavy export trucks loading dried fruit; visitor arrivals; expansive 20+ hectare open farm footprint.

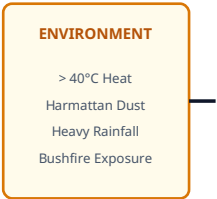
Engineering Impact: Access control must accommodate rapid batch verification without bottlenecking operations. High open-space visibility is mandatory to detect unauthorized movements across farm zones without hindering agricultural machinery.

D. Security Context (Asset Protection & Chain-of-Custody)

Key Factors: High market value of processed dried mangoes; expensive processing machinery and solar power infrastructure; physical safety of staff; maintaining uncompromised organic export certification.

Engineering Impact: Requires multi-layered perimeter defense (Defense-in-Depth) ensuring zero unauthorized access into high-hygiene drying facilities and sealed export containers.

INTEGRATED CONTEXT INTERACTION MODEL



3. COMPREHENSIVE ASSET IDENTIFICATION

To design an effective security system, every asset within the farm ecosystem must be identified, categorized, and evaluated for criticality. Assets are grouped into four core categories.

PHYSICAL ASSETS (Infrastructure & Goods)

- **Finished Inventory:** Dried mango packages ready for export (Highest financial & food safety value).
- **Processing & Drying Facility:** Hygienic ovens, slicing lines, packaging zones.
- **Off-Grid Solar Power Plant:** Photovoltaic panels, inverter station, lithium battery bank.
- **Perimeter Barriers:** Outer security fences, structural access gates, vehicle barriers.
- **Farm Machinery & Vehicles:** Tractors, harvest haulers, maintenance tools.

HUMAN ASSETS (People & Safety)

- **Permanent Facility Staff:** Plant engineers, quality control technicians, administrative staff.
- **Seasonal Agricultural Workers:** Large seasonal labor force performing harvest and initial sorting.
- **Logistics Drivers & Visitors:** External truck drivers, auditors, organic certifiers, contractors.
- **Security Personnel:** On-site guard force operating access points and performing patrols.

OPERATIONAL ASSETS (Processes & Continuity)

- **Drying & Processing Workflow:** Continuous temperature-controlled drying cycle; stoppage causes fruit spoilage.
- **Supply Chain Traceability:** Organic chain-of-custody documentation required for German import standards.
- **Logistics & Loading Schedule:** Timely dispatch of containerized fruit to port/airport.
- **Visitor & Labor Access Management:** Daily check-in, identity verification, and zone authorization.

DIGITAL ASSETS (Data & Systems)

- **Access & Visitor Registers:** Digital logs of personnel movements and vehicle entry/exit.
- **Video Surveillance Footage:** Local CCTV recordings for incident analysis and export audit compliance.
- **Intrusion Alarm System State:** Sensor states, perimeter breach alerts, system status logs.
- **Power System Diagnostics:** Solar charge control data, battery temperature, and health telemetry.

ASSET CRITICALITY HIERARCHY

Level	Asset Category	Primary Impact if Compromised
Critical	Processed Mango Inventory & Solar Power Bank	Immediate total financial loss, delivery failure to Germany, facility shutdown.
High	Human Safety & Processing Plant Equipment	Operational halts, legal liability, costly equipment replacement timelines.
Medium	Digital Logs & Farm Machinery	Temporary audit friction, localized agricultural delay.

4. CATEGORIZED RISK CLASSIFICATION

Security failures occur when systems are designed for narrow threat models while ignoring broader operational hazards. Risks acting upon the facility are categorized into six structural vectors.

1. Human Security Risks (Intentional & Internal Threats)

- **Perimeter Theft & Intrusion:** Unauthorized external entry to steal dried fruit packages or high-value solar components.
- **Vandalism & Sabotage:** Cut fences, damaged camera lenses, or malicious tampering with facility equipment.
- **Insider Misuse:** Theft of product by employees or collusion between guards and external parties during night shifts.

2. Climate & Environmental Risks (Sahelian Operating Stressors)

- **Extreme Thermal Degradation:** Ambient heat > 40°C causing outdoor electronics over-heating and battery degradation.
- **Harmattan Dust Accumulation:** Fine dust settling on optical CCTV lenses, motion sensors, and solar arrays.
- **Monsoon Torrential Rain & Wind:** Flooding near perimeter foundations; high winds misaligning PTZ camera mounts.

3. Infrastructure Vulnerability Risks (System Support Failures)

- **Complete Off-Grid Power Loss:** Battery depletion during cloudy periods causing security camera blackouts.
- **Telecommunications Network Failure:** Loss of cellular internet preventing remote alert transmission to management.
- **Hardware Wear & Tear:** Premature degradation of mechanical locks, gate hinges, and sensors due to dust.

4. Operational & Human Error Risks (Procedural Vulnerabilities)

- **Access Control Bottlenecks:** Guard errors during peak seasonal morning shifts leading to unverified entry.
- **Poor Maintenance Compliance:** Neglecting sensor cleaning, vegetation trimming along fences, or battery checks.
- **Inadequate Visitor Registration:** Unverified contractors entering sensitive processing areas.

5. Site Environmental Hazards (Surrounding Ecosystem Risks)

- **Bushfire Encroachment:** Dry brush fires along perimeter fences threatening physical structural integrity.
- **Wildlife & Vegetation Overgrowth:** Animals triggering false motion alarms; dense brush obstructing camera line-of-sight.

6. Supply Chain & Traceability Risks (Export Logistics Threats)

- **Loading Dock Theft:** Product diversion during container loading operations.
- **Cargo Tampering & Substitution:** Breach of export seals introducing non-organic or contaminated fruit prior to dispatch.

5. RISK ASSESSMENT & MATRIX

To prioritize engineering resources, identified risks were evaluated using a standard 5×5 Likelihood vs. Impact matrix adapted to the rural Burkina Faso agricultural operational context.

RISK EVALUATIVE MATRIX (5 × 5)

	Negligible (1)	Minor (2)	Moderate (3)	Major (4)	Catastrophic (5)
Almost Certain (5)	Dust Accumulation	Thermal Heat Stress	Access Gate Delays	Product Theft (Night)	Solar System Theft
Likely (4)	Minor Vandalism	Vegetation Obstruction	Network Blackout	Power Depletion	Contamination / Sabotage
Possible (3)	Animal False Alarm	Visitor Log Error	Insider Product Diversion	Perimeter Fence Cut	Bushfire Encroachment
Unlikely (2)	Tool Loss	Camera Tilt (Wind)	Loading Dock Confusion	Vehicle Barrier Failure	Export Seal Tampering
Rare (1)	Negligible Lapses	Water Intrusion	Sensor Miscalibration	Armed Robbery Threat	Container Contamination

PRIORITIZED ENGINEERING RISK REGISTER

ID	Risk Event	Likelihood	Impact	Priority	Proposed Engineering Mitigation
R-01	Solar Panel & Battery Bank Theft	Almost Certain	Catastrophic	CRITICAL	Inner security zone enclosure, vibration sensors, tamper-proof mounting bolts, dedicated CCTV coverage.
R-02	Dried Mango Finished Cargo Theft	Almost Certain	Major	CRITICAL	Reinforced warehouse access controls, dual-custody locking, thermal detection along storage perimeter.
R-03	Thermal Camera Failure (>40°C)	Almost Certain	Minor	HIGH	Specify IP67/IK10 outdoor enclosures with sunshields and passive thermal dissipation fins.
R-04	Harmattan Dust Sensor Blindness	Almost Certain	Negligible	MEDIUM	Scheduled physical lens maintenance, dust-resistant seals, sealed optical enclosures.
R-05	Perimeter Breach / Fence Cutting	Possible	Major	HIGH	Anti-climb rigid mesh wire, clear inner buffer zone, perimeter zone alarm integration.
R-06	Perimeter Bushfire Damage	Possible	Catastrophic	CRITICAL	10-meter cleared firebreak perimeter zone, non-combustible metallic fencing elements.
R-07	Off-Grid Power Depletion	Likely	Major	CRITICAL	Dedicated 72-hour security battery backup completely isolated from plant operations.

6. RISK → ENGINEERING IMPLICATION MAPPING

This section represents the analytical core of Week 3. Every identified operational risk is directly translated into a functional engineering requirement and a concrete design decision for subsequent system design.

TRACEABILITY FLOW MATRIX: Risk Identification → Engineering Implication → Future Design Decision

EXTREME HEAT (>40°C)	→	Outdoor Industrial Equipment Requirement	→	High-temperature rated cameras (-20°C to +65°C) with sunshield visors & passive cooling.
HARMATTAN DUST	→	Ingress & Optical Protection Needs	→	IP67 weather-resistant enclosures, hydrophobic glass coatings, monthly lens maintenance protocol.
POWER OUTAGES	→	System Continuity & Power Autonomy	→	Dedicated off-grid solar + LiFePO4 battery storage providing 72 hours continuous operation.
UNAUTHORIZED ACCESS	→	Access Channel Control & Bottlenecking	→	Single controlled main entry gate, turnstiles for workers, visitor registration guardhouse.
PRODUCT & ASSET THEFT	→	Multi-Layered Asset Protection Zone	→	Restricted storage zones, anti-climb rigid mesh perimeter fence, infrared night surveillance.
BUSHFIRE HAZARD	→	Emergency Preparedness & Fire Defense	→	10m vegetation-free firebreak perimeter, metal posts, flame-retardant cabling conduits.
UNSTABLE NETWORK	→	Localized Data Reliability	→	Edge storage (SD cards in cameras + local NVR) with asynchronous sync when cloud connection returns.

Engineering Justification: By establishing this direct line of traceability, every component specified in Week 5 (Zoning) and Week 6 (Physical Design) will possess a clear justification linked back to an analyzed operational risk.

7. WEEK 3 DELIVERABLES SUMMARY

During Week 3, three primary technical deliverables were developed to support the engineering progression of the Security & Fencing System.

Deliverable 1: Week 3 Engineering PDF Report (This Document)

- **Description:** Formal systems engineering document presenting environmental context, asset criticalities, risk matrices, and engineering implications.
- **Target Audience:** Excellence Program mentors, project directorate, and lead site engineers.
- **Role in Project:** Serves as the authoritative baseline for all physical security design decisions.

Deliverable 2: Prioritized Engineering Risk Register

- **Description:** Living, structured database cataloging 15+ site-specific operational, environmental, and human risks categorized by Likelihood and Impact.
- **Key Feature:** Explicitly links environmental stressors (Harmattan dust, 45°C heat) to hardware protection classes (IP67, IK10).
- **Role in Project:** Guides procurement specifications and security zoning boundaries.

Deliverable 3: Miro Interactive Engineering Board

- **Description:** Visual system mapping board illustrating real-time connectivity between system elements.
- **Traceability Path:**

Identified Assets → Risk Categories → Risk Priorities → Engineering Decisions → Week 4 Input
- **Role in Project:** Enables collaborative design review and visual feedback validation.

TRANSITION TO WEEK 4: FARM FUNCTIONAL ANALYSIS

Risk and asset analysis is not the end goal of security engineering; it is the prerequisite. With assets cataloged and operational risks prioritized, Week 4 will transition into the **Farm Functional Analysis**.

Week 4 will map how workers, trucks, visitors, and produce physically move through the facility, establishing security zones and functional workflows based on the risks identified this week.

PROJECT ROADMAP ALIGNMENT

Week 1 Brainstorming Lessons Learned	Week 2 Design Strategy Methodology	Week 3 (Current) Risk & Asset Analysis Context & Threats	Week 4 Functional Analysis Workflows	Week 5 Security Zoning Perimeters	Week 6 Physical Design Final Spec
--	--	--	--	---	---

Conclusion: Every future engineering choice—from fence wire gauge to camera sensor IP ratings—will be directly justified by the rigorous environmental and threat baseline established in Weeks 1 through 3.